 I S B N：978-986-02-7580-3

On the Antimagic Labeling of Star Forests *

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Abstract

In this paper we investigate the antimagic labeling of star forests and show that a S_1-free star forest $\cup_{i=1}^{r} S_{k_i}$ where $r \geq 2$ which contains at most one S_2 is antimagic. We also prove that a star forest mS_2 is antimagic if and only if $m = 1$. In addition, for $n \geq 14$ we show that a star forest $mS_2 \cup S_n$ is antimagic if and only if $m \leq 2n + 1$.

Keywords: antimagic labeling, star forest

1 Introduction

All the graphs considered in this paper are finite, simple and without any isolated vertex. For a graph $G(V(G), E(G))$ with p vertices and q edges, an *edge labeling* of G is a bijection $f : E(G) \to \{1, 2, \cdots, q\}$ and the induced *vertex sum* $f^+ : V(G) \to N$ of the edge labeling f is a function given by $f^+(u) = \sum_{uv \in E(G)} f(uv)$ for all $u \in V(G)$. And $f^+(u)$ is called the *vertex sum* of u. G is called *antimagic* if there exists an edge labeling of G such that the induced vertex sum of the edge labeling is injective. And we call the edge labeling of G an *antimagic labeling* of G. Hartsfield and Ringel [3] introduced the concept of the antimagic labeling of graphs first. They showed that paths, cycles, complete graphs $K_n (n \geq 3)$ are antimagic and conjectured that all connected graphs except K_2 are antimagic, which remains to be solved. Alon et al [1] proved that this conjecture is true for dense graphs, they showed that all graphs with p vertices ($p \geq 4$) and minimum degree $\Omega(\log p)$ are antimagic. They also proved that complete partite graphs besides K_2 are antimagic. Let P_n denote a path of order n and C_n denote a cycle of order n. Wang and Hsiao [4] constructed antimagic graphs through Cartesian products. They showed that $P_m \times P_n (m \geq n \geq 2)$, $H \times P_m$ and $H \times C_n$ where H is a d-regular graph ($d \geq 1, m \geq 2, n \geq 3$) are antimagic. For more graphs which are known antimagic, please see [2]. For $\ell \in N$, a *star* S_ℓ is the complete bipartite graph $K_{1,\ell}$. For $\ell \geq 2$, the vertex of degree ℓ in S_ℓ is called the *center* of S_ℓ and any vertex of degree 1 in S_ℓ is called an *endvertex* of S_ℓ. A *star forest* is a graph consists of r number of disjoint stars, where $r \geq 1$. We denote a star forest of r stars by $\cup_{i=1}^{r} S_{k_i}$. If $k_i = t$ for $i = 1, 2, \cdots, r$ where $t \geq 1$, then we simply write $\cup_{i=1}^{r} S_{k_i}$ by rS_t. In this article we investigate the antimagicness of star forests. It is obvious that any star forest containing S_1 is not antimagic. Let $\cup_{i=1}^{r} S_{k_i}$ be a star forest and f be an edge labeling of $\cup_{i=1}^{r} S_{k_i}$ and f^+ be the induced vertex sum of f. For $i = 1, 2, \cdots, r$, let $E(S_{k_i}) = \{\epsilon_{i,1}, \epsilon_{i,2}, \cdots, \epsilon_{i,k_i}\}$. Suppose $f(\epsilon_{i,j}) = e_{i,j}$ for $j = 1, 2, \cdots, k_i$ and $i = 1, 2, \cdots, r$. It is easy to see that $f^+(V(S_{k_i})) = \{e_{i,1}, e_{i,2}, \cdots, e_{i,k_i}, \sum_{j=1}^{k_i} e_{i,j}\}$ where each $e_{i,j}$ is the vertex sum of an endvertex of S_{k_i} and $\sum_{j=1}^{k_i} e_{i,j}$ is the vertex sum of the center of S_{k_i}. Since $f^+(V(S_{k_i}))$ is uniquely determined by f for $i = 1, 2, \cdots, r$, then for the convenience of our discussions below, we simply write the edge labels on $E(S_{k_i})$ and the vertex sums on $V(S_{k_i})$ by

$$S_{k_i} \to \begin{pmatrix} e_{i,1}, e_{i,2}, \cdots, e_{i,k_i} \\ \sum_{j=1}^{k_i} e_{i,j} \end{pmatrix}$$

*This research is supported by NSC of R.O.C. under grant NSC 99-2115-M-008-003-MY2

where the upper row $e_{i,1}, e_{i,2}, \cdots, e_{i,k_i}$ in the bracket denotes both the edge labels on $E(S_{k_i})$ and the vertex sums of the endvertices of S_{k_i}, and the lower integer $\sum_{j=1}^{k_i} e_{i,j}$ in the bracket denotes the vertex sum of the center of S_{k_i}. f is an antimagic labeling of $\cup_{i=1}^{r} S_{k_i}$ if the induced vertex sum f^+ defined on $V(\cup_{i=1}^{r} S_{k_i})$ is injective. Hence if the integers in brackets

$$\begin{pmatrix} e_{1,1}, e_{1,2}, \cdots, e_{1,k_1} \\ \sum_{j=1}^{k_1} e_{1,j} \end{pmatrix}, \begin{pmatrix} e_{2,1}, e_{2,2}, \cdots, e_{2,k_2} \\ \sum_{j=1}^{k_2} e_{2,j} \end{pmatrix}, \cdots,$$

$\begin{pmatrix} e_{r,1}, e_{r,2}, \cdots, e_{r,k_r} \\ \sum_{j=1}^{k_r} e_{r,j} \end{pmatrix}$ are all distinct, then f is an antimagic labeling of $\cup_{i=1}^{r} S_{k_i}$. Note that since f is an edge labeling of $\cup_{i=1}^{r} S_{k_i}$, we have

$$\cup_{i=1}^{r} \cup_{j=1}^{k_i} \{e_{i,j}\} = \{1, 2, \cdots, \sum_{i=1}^{r} k_i\}.$$

A star forest consists of a single star S_ℓ is antimagic whenever $\ell \geq 2$ for $S_\ell \to \begin{pmatrix} 1, 2, \cdots, \ell \\ \frac{(1+\ell)\ell}{2} \end{pmatrix}$, which is also mentioned in [2].

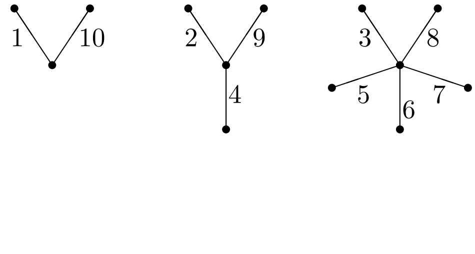
2 Main Results

We begin this section with the following theorem.

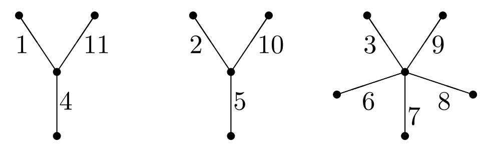
Theorem 2.1 *A star forest $\cup_{i=1}^{r} S_{k_i}$ with $k_1 \geq 2$ and $k_i \geq 3$ for $i = 2, 3, \cdots, r$ where $r \geq 2$ is antimagic.*

Proof. We first give two examples to show our way for giving an antimagic labeling for a S_1-free star forest which contains at most one S_2. Note that in these examples every number beside an edge is the label of this edge.

Example 1: $S_2 \cup S_3 \cup S_5$; a S_1-free star forest which contains one S_2.



Example 2: $2S_3 \cup S_5$; a S_1-free star forest which contains no S_2.



Now, we start our proof. Without loss of generality, let $2 \leq k_1 \leq k_2 \leq k_3 \leq \cdots \leq k_r$ and $k_2 \geq 3$. For $i = 1, 2, \cdots, r$, let x_i be the center of S_{k_i} and $y_{i,1}, y_{i,2}, \cdots, y_{i,k_i}$ be the endvertices of S_{k_i} and $\epsilon_{i,j} = x_i y_{i,j}$ for $j = 1, 2, \cdots, k_i$. Then $E(\cup_{i=1}^{r} S_{k_i}) = \cup_{i=1}^{r} \cup_{j=1}^{k_i} \{\epsilon_{i,j}\}$. Suppose $e = \sum_{i=1}^{r} k_i$, then $|E(\cup_{i=1}^{r} S_{k_i})| = e$. Now we give a function f defined on $E(\cup_{i=1}^{r} S_{k_i})$ by

$$\begin{cases} f(\epsilon_{i,1}) = i, & 1 \leq i \leq r, \\ f(\epsilon_{i,2}) = e - i + 1, & 1 \leq i \leq r, \\ f(\epsilon_{1,j}) = r + j - 2, & 3 \leq j \leq k_1, \\ f(\epsilon_{i,j}) = r + \sum_{t=1}^{i-1}(k_t - 2) + j - 2, & 3 \leq j \leq k_i, \\ & 2 \leq i \leq r. \end{cases}$$

It is easy to see that f is an edge labeling of $\cup_{i=1}^{r} S_{k_i}$. Now we claim $1 + e \leq f^+(x_1) < f^+(x_2)$. We distinguish the following two cases:
Case 1. $k_1 = 2$.

$$\begin{aligned} f^+(x_2) &= \sum_{j=1}^{k_2} f(\epsilon_{2,j}) \\ &= f(\epsilon_{2,1}) + f(\epsilon_{2,2}) + \sum_{j=3}^{k_2} f(\epsilon_{2,j}) \\ &> 1 + e \qquad (\text{ for } f(\epsilon_{2,3}) > 0 \text{ }) \\ &= f(\epsilon_{1,1}) + f(\epsilon_{1,2}) \\ &= f^+(x_1). \qquad (\text{ for } k_1 = 2 \text{ }) \end{aligned}$$

Case 2. $k_1 \geq 3$.

$$\begin{aligned} f^+(x_2) &= \sum_{j=1}^{k_2} f(\epsilon_{2,j}) \\ &= f(\epsilon_{2,1}) + f(\epsilon_{2,2}) + \sum_{j=3}^{k_2} f(\epsilon_{2,j}) \\ &\geq 1 + e + \sum_{j=3}^{k_1} f(\epsilon_{2,j}) \quad (\text{ for } k_2 \geq k_1 \text{ }) \\ &> 1 + e + \sum_{j=3}^{k_1} f(\epsilon_{1,j}) \\ &\quad (\text{ for } f(\epsilon_{2,j}) > f(\epsilon_{1,j}), j = 3, \cdots, k_1 \text{ }) \\ &= f^+(x_1). \end{aligned}$$

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By Case 1 and Case 2, we have $1+e \leq f^+(x_1) < f^+(x_2)$.

For $2 \leq i \leq r-1$, consider the vertex sums for the centers x_{i+1} and x_i, we have

$$\begin{aligned}
f^+(x_{i+1}) &= \sum_{j=1}^{k_{i+1}} f(\epsilon_{i+1,j}) \\
&= f(\epsilon_{i+1,1}) + f(\epsilon_{i+1,2}) + \sum_{j=3}^{k_{i+1}} f(\epsilon_{i+1,j}) \\
&\geq 1+e+\sum_{j=3}^{k_i} f(\epsilon_{i+1,j}) \\
&\qquad (\text{ for } k_{i+1} \geq k_i) \\
&> 1+e+\sum_{j=3}^{k_i} f(\epsilon_{i,j}) \\
&\qquad (\text{ for } f(\epsilon_{i+1,j}) > f(\epsilon_{i,j}), \\
&\qquad\quad j = 3, \cdots, k_i) \\
&= f^+(x_i).
\end{aligned}$$

Then we have $1+e \leq f^+(x_1) < f^+(x_2) < f^+(x_3) < \cdots < f^+(x_r)$ and $\cup_{i=1}^r \cup_{j=1}^{k_i} \{f^+(y_{i,j})\} = \{1,2,\cdots,e\}$. Thus $f^+ : V(\cup_{i=1}^r S_{k_i}) \to N$ is injective and we conclude that $\cup_{i=1}^r S_{k_i}$ is antimagic. □

Theorem 2.2 *The star forests mS_2 is antimagic if and only if $m=1$.*

Proof. $\Rightarrow$) Suppose mS_2 is antimagic and $m \geq 2$. Let $S_i' = S_2$ for $i = 1,2,\cdots,m$. Then $mS_2 = \cup_{i=1}^m S_i'$. For $i = 1,2,\cdots,m$, let x_i be the center of S_i' and $y_{i,1}, y_{i,2}$ be the endvertices of S_i'. Then $V(\cup_{i=1}^m S_i') = \cup_{i=1}^m \{x_i, y_{i,1}, y_{i,2}\}$ and $E(\cup_{i=1}^m S_i') = \cup_{i=1}^m \{x_i y_{i,1}, x_i y_{i,2}\}$. By assumption, the star forest mS_2 is antimagic, then there exists an edge labeling $f : \cup_{i=1}^m \{x_i y_{i,1}, x_i y_{i,2}\} \to \{1,2,\cdots,2m\}$ such that all the induced vertex sums $f^+(x_i), f^+(y_{i,j})$ where $i = 1,2,\cdots m$ and $j = 1,2$ are pairwise distinct. Since vertex sums $f^+(y_{i,j})$ where $i = 1,2,\cdots m$ and $j = 1,2$ receive the values $1,2,\cdots,2m$, we have $f^+(x_i) \geq 2m+1$ for $i = 1,2,\cdots m$. Without loss of generality, assume that $f(x_1y_{1,1}) = 1$, then we have $f(x_1y_{1,2}) = 2m$. For if not, i.e. $f(x_1y_{1,2}) \leq 2m-1$, then $f^+(x_1) = f(x_1y_{1,1}) + f(x_1y_{1,2}) \leq 1+(2m-1) = 2m$. Which contradicts that $f^+(x_1) \geq 2m+1$. Now we have $f^+(x_1) = 2m+1$ and then $f^+(x_i) \geq 2m+2$ for $i = 2,3,\cdots m$. Again, without loss of generality, assume that $f(x_2y_{2,1}) = 2$. Since $f^+(x_2) = f(x_2y_{2,1}) + f(x_2y_{2,2})$, then $f(x_2y_{2,2}) = f^+(x_2) - f(x_2y_{2,1}) \geq (2m+2)-2 = 2m$. Which is a contradiction for $f(x_2y_{2,2}) \leq 2m-1$. From the paradox above, we conclude that if mS_2 is antimagic then $m=1$.

$\Leftarrow$) S_2 is antimagic for $S_2 \to \begin{pmatrix} 1,2 \\ 3 \end{pmatrix}$. □

Lemma 2.3 *Let n be an integer ≥ 3. If $mS_2 \cup S_n$ is antimagic, then*

$$m \leq \min\{2n+1, \frac{2n-5+\sqrt{8n^2-24n+17}}{2}\}.$$

Proof. Let $x_1, x_2, \cdots, x_m, x_{m+1}$ be the centers of $mS_2 \cup S_n$ and $y_1, y_2, \cdots, y_{2m+n}$ be the endvertices of $mS_2 \cup S_n$. By assumption, $mS_2 \cup S_n$ is antimagic, then there exists an edge labeling $f : E(mS_2 \cup S_n) \to \{1,2,\cdots,2m+n\}$ such that all the induced vertex sums $f^+(x_1), f^+(x_2), \cdots, f^+(x_{m+1}), f^+(y_1), f^+(y_2), \cdots, f^+(y_{2m+n})$ are pairwise distinct. For $f^+(y_1), f^+(y_2), \cdots, f^+(y_{2m+n})$ receive the values $1,2,\cdots,2m+n$, we have $f^+(x_i) \geq 2m+n+1$ for $i = 1,2,\cdots,m+1$. Hence

$$\begin{aligned}
& f^+(x_1) + f^+(x_2) + \cdots + f^+(x_{m+1}) \\
& \geq \sum_{i=1}^{m+1} (2m+n+i) \\
& = \frac{1}{2}(5m+2n+2)(m+1).
\end{aligned}$$

And it is clear that

$$\begin{aligned}
& f^+(x_1) + f^+(x_2) + \cdots + f^+(x_{m+1}) \\
& = 1+2+\cdots+(2m+n) \\
& = \frac{1}{2}(2m+n+1)(2m+n).
\end{aligned}$$

Then we have

$$(2m+n+1)(2m+n) \geq (5m+2n+2)(m+1),$$

$$4m^2+(4n+2)m+n^2+n \geq 5m^2+(2n+7)m+2n+2,$$

$$m^2+(5-2n)m+2+n-n^2 \leq 0,$$

$$m \leq \frac{2n-5+\sqrt{8n^2-24n+17}}{2}.$$

Next, we show that $m \leq 2n+1$. Consider the vertex sums $f^+(x_1), f^+(x_2), \cdots, f^+(x_m)$ of the centers of all S_2 in $mS_2 \cup S_n$. Since $\sum_{i=1}^m f^+(x_i)$ can not exceed the sum of the largest $2m$ edge labels $2m+n, 2m+n-1, 2m+n-2, \cdots, n+1$, we have

$$\sum_{i=1}^m (2m+n+i) \leq \sum_{i=1}^m f^+(x_i) \leq \sum_{i=0}^{2m-1} (2m+n-i).$$

 ISBN：978-986-02-7580-3

Then

$$\begin{aligned}\frac{1}{2}(5m+2n+1)m &\le \frac{1}{2}(2m+2n+1)2m,\\ m &\le 2n+1.\end{aligned}$$

Hence we have

$$m \le \min\{2n+1, \frac{2n-5+\sqrt{8n^2-24n+17}}{2}\}.$$

□

Remark 2.4 *For $n \in N$,*

$$\min\{2n+1, \frac{2n-5+\sqrt{8n^2-24n+17}}{2}\}$$
$$= \begin{cases} \frac{2n-5+\sqrt{8n^2-24n+17}}{2} & if \quad n \le 13,\\ 2n+1 & if \quad n \ge 14.\end{cases}$$

Proof. Suppose $n \in N$. Then $8n^2-24n+17 > 0$. And since

$$\begin{aligned}&\frac{2n-5+\sqrt{8n^2-24n+17}}{2} > 2n+1\\ \Leftrightarrow\ & 2n-5+\sqrt{8n^2-24n+17} > 4n+2\\ \Leftrightarrow\ & \sqrt{8n^2-24n+17} > 2n+7\\ \Leftrightarrow\ & 8n^2-24n+17 > (2n+7)^2\\ \Leftrightarrow\ & n^2-13n-8 > 0.\end{aligned}$$

We conclude that if $n \ge 14$ then $\frac{2n-5+\sqrt{8n^2-24n+17}}{2} > 2n+1$ and if $n \le 13$ then $2n+1 > \frac{2n-5+\sqrt{8n^2-24n+17}}{2}$. Hence Remark 2.4 holds. □

Lemma 2.5 *For $n \ge 10$ and $m \le n+1$, $mS_2 \cup S_n$ is antimagic.*

Proof. Let $S'_i = S_2$ for $i = 1, 2, \cdots, m$. Then $mS_2 \cup S_n = \cup_{i=1}^m S'_i \cup S_n$. Let x_i be the center of S'_i for $i = 1, 2, \cdots, m$ and x_{m+1} be the center of S_n. Let f be a function defined on $E(\cup_{i=1}^m S'_i \cup S_n)$ with the following restricted images on stars S_n and $S'_i, i = 1, 2, \cdots, m$.

$$\begin{cases} f(E(S_n)) = & \{1, 2, \cdots n\},\\ f(E(S'_i)) = & \{n+i, n+m+i\}\\ & for\ i = 1, 2, \cdots, m.\end{cases}$$

Claim: any f satisfying above conditions is an edge labeling of $\cup_{i=1}^m S'_i \cup S_n$.
For

$$\begin{aligned}&\{1, 2, \cdots, 2m+n\}\\ =\ & \{1, 2, \cdots, n\} \cup \{n+1, n+2, \cdots, n+m\}\\ & \cup\{n+m+1, n+m+2, \cdots, n+2m\}\\ =\ & \{1, 2, \cdots, n\} \cup (\cup_{i=1}^m\{n+i\})\\ & \cup(\cup_{i=1}^m\{n+m+i\})\\ =\ & \{1, 2, \cdots, n\} \cup (\cup_{i=1}^m\{n+i, n+m+i\})\\ =\ & f(E(S_n)) \cup (\cup_{i=1}^m f(E(S'_i)))\\ =\ & f(E(\cup_{i=1}^m S'_i \cup S_n)),\end{aligned}$$

and $|E(\cup_{i=1}^m S'_i \cup S_n)| = 2m+n$. We see that such an $f : E(\cup_{i=1}^m S'_i \cup S_n) \to \{1, 2, \cdots, 2m+n\}$ is a bijection, hence f is an edge labeling of $\cup_{i=1}^m S'_i \cup S_n$. Then the induced vertex sum f^+ of f is

$$\begin{cases} S_n & \to \begin{pmatrix} 1, 2, \cdots, n\\ \frac{(1+n)n}{2}\end{pmatrix},\\ S'_i & \to \begin{pmatrix} n+i, n+m+i\\ 2n+m+2i\end{pmatrix}\\ & for\ i = 1, 2, \cdots, m.\end{cases}$$

Next we show that f^+ is injective. We see that $f^+(x_{m+1}) = \frac{(1+n)n}{2}$ and $f^+(x_i) = 2n+m+2i$ for $i = 1, 2, \cdots, m$. Then

$$\begin{aligned}& f^+(x_1) - (2m+n)\\ =\ & (2n+m+2) - (2m+n)\\ =\ & n-m+2 > 0 \quad \text{for} \quad m \le n+1,\end{aligned}$$

and

$$\begin{aligned}& f^+(x_{m+1}) - f^+(x_m)\\ =\ & \frac{(1+n)n}{2} - (3m+2n)\\ \ge\ & \frac{n^2+n}{2} - 3(n+1) - 2n \quad \text{for } m \le n+1\\ =\ & \frac{n^2-9n-6}{2} > 0 \quad \text{if} \quad n \ge 10.\end{aligned}$$

Then we have the vertex sums of the centers in $\cup_{i=1}^m S'_i \cup S_n$ are in order of $f^+(x_{m+1}) > f^+(x_m) > f^+(x_{m-1}) > \cdots > f^+(x_2) > f^+(x_1) > 2m+n$ if $n \ge 10$ and all the endvertices of $\cup_{i=1}^m S'_i \cup S_n$ receive vertex sums $1, 2, \cdots, 2m+n$. Then $f^+ : V(\cup_{i=1}^m S'_i \cup S_n) \to N$ is injective. Hence $\cup_{i=1}^m S'_i \cup S_n = mS_2 \cup S_n$ is antimagic. □

Let E denote the set of positive even integers and O denote the set of positive odd integers. Then $N = E \cup O$. If A, B are two sets of integers and $A \subset E, B \subset O$ (or $A \subset O, B \subset E$), then we say that A and B are in different parties of N.

Lemma 2.6 *For $n \ge 14$ and $n+2 \le m \le 2n+1$, $mS_2 \cup S_n$ is antimagic.*

Proof. Let $S'_i = S_2$ for $i = 1, 2, \cdots, m$. Then $mS_2 \cup S_n = \cup_{i=1}^m S'_i \cup S_n$. Let x_i be the center of

 ISBN：978-986-02-7580-3

S_i' for $i = 1, 2, \cdots, m$ and x_{m+1} be the center of S_n. We distinguish two cases and in each case we give a function defined on $E(\cup_{i=1}^m S_i' \cup S_n)$ which has the following restricted images on stars S_n and $S_i', i = 1, 2, \cdots, m$.

Case 1. m is odd and $n+2 \le m \le 2n+1$. Let f be a function defined on $E(\cup_{i=1}^m S_i' \cup S_n)$ satisfying

$$\begin{cases} f(E(S_n)) = & \{1, 2, \cdots n\}, \\ f(E(S_i')) = & \{n+i, 2m-1+i\} \\ & for\ i = 1, 2, \cdots, n+1, \\ f(E(S_i')) = & \{n+i, m-1+i\} \\ & for\ i = n+2, n+3, \cdots, m. \end{cases}$$

Claim: any f satisfying above conditions is an edge labeling of $\cup_{i=1}^m S_i' \cup S_n$. First, we see that

$\cup_{i=1}^{n+1}\{n+i\} = \{n+1, n+2, \cdots, 2n+1\}$,

$\cup_{i=n+2}^{m}\{n+i\} = \{2n+2, 2n+3, \cdots, m+n\}$,

$\cup_{i=n+2}^{m}\{m-1+i\} = \{m+n+1, m+n+2, \cdots, 2m-1\}$,

$\cup_{i=1}^{n+1}\{2m-1+i\} = \{2m, 2m+1, \cdots, 2m+n\}$

are disjoint sets. And

$$\begin{aligned} & \{1, 2, \cdots, 2m+n\} \\ = & \{1, 2, \cdots, n\} \cup \{n+1, n+2, \cdots, 2n+1\} \\ & \cup\{2n+2, 2n+3, \cdots, m+n\} \\ & \cup\{m+n+1, m+n+2, \cdots, 2m-1\} \\ & \cup\{2m, 2m+1, \cdots, 2m+n\} \\ = & \{1, 2, \cdots, n\} \cup (\cup_{i=1}^{n+1}\{n+i\}) \\ & \cup(\cup_{i=n+2}^{m}\{n+i\}) \cup (\cup_{i=n+2}^{m}\{m-1+i\}) \\ & \cup(\cup_{i=1}^{n+1}\{2m-1+i\}) \\ = & \{1, 2, \cdots, n\} \cup (\cup_{i=1}^{n+1}\{n+i, 2m-1+i\}) \\ & \cup(\cup_{i=n+2}^{m}\{n+i, m-1+i\}) \\ = & f(E(S_n)) \cup (\cup_{i=1}^{n+1} f(E(S_i'))) \\ & \cup(\cup_{i=n+2}^{m} f(E(S_i'))) \\ = & \cup_{i=1}^{m} f(E(S_i')) \cup f(E(S_n)) \\ = & f(E(\cup_{i=1}^m S_i' \cup S_n)). \end{aligned}$$

And $|E(\cup_{i=1}^m S_i' \cup S_n)| = 2m+n$. We have $f : E(\cup_{i=1}^m S_i' \cup S_n) \to \{1, 2, \cdots, 2m+n\}$ is a bijection. Hence f is an edge labeling of $\cup_{i=1}^m S_i' \cup S_n$. Then the induced vertex sum f^+ of f is

$$\begin{cases} S_n & \to \begin{pmatrix} 1, 2, \cdots, n \\ \frac{(1+n)n}{2} \end{pmatrix}, \\ S_i' & \to \begin{pmatrix} n+i, 2m-1+i \\ 2m+n-1+2i \end{pmatrix} \\ & \quad for\ i = 1, 2, \cdots, n+1, \\ S_i' & \to \begin{pmatrix} n+i, m-1+i \\ m+n-1+2i \end{pmatrix} \\ & \quad for\ i = n+2, n+3, \cdots, m. \end{cases}$$

Now we show that f^+ is injective. From above, we see that all the endvertices of $\cup_{i=1}^m S_i' \cup S_n$ receive vertex sums $1, 2, \cdots, 2m+n$. Let

$$\begin{aligned} A &= \{f^+(x_1), f^+(x_2), \cdots, f^+(x_{n+1})\} \\ &= \cup_{i=1}^{n+1}\{2m+n-1+2i\}, \\ B &= \{f^+(x_{n+2}), f^+(x_{n+3}), \cdots, f^+(x_m)\} \\ &= \cup_{i=n+2}^{m}\{m+n-1+2i\}. \end{aligned}$$

We have $A \subset O$ or $A \subset E$ and $B \subset O$ or $B \subset E$. And since

$$\begin{aligned} & f^+(x_{n+2}) - f^+(x_1) \\ = & (m+3n+3) - (2m+n+1) \\ = & 2n-m+2 \end{aligned}$$

is odd and positive for m is odd and $m \le 2n+1$. Then we have A and B are in different parties of N (i.e., if $A \subset O$ then $B \subset E$ or if $A \subset E$ then $B \subset O$) and

$$f^+(x_m) > \cdots > f^+(x_{n+3}) > f^+(x_{n+2}) > 2m+n,$$

$$f^+(x_{n+1}) > \cdots > f^+(x_2) > f^+(x_1) > 2m+n.$$

i.e. all centers $x_1, x_2, \cdots, x_m$ of $S_1', S_2', \cdots, S_m'$ receive different vertex sums $> 2m+n$. Now, we consider the vertex sum $f^+(x_{m+1})$ for the center x_{m+1} of star S_n.

Since

$$\begin{aligned} & f^+(x_{m+1}) - f^+(x_{n+1}) \\ = & \frac{(1+n)n}{2} - (2m+3n+1) \\ \ge & \frac{n^2+n}{2} - 2(2n+1) - 3n - 1 \text{ for } m \le 2n+1 \\ = & \frac{n^2-13n-6}{2} > 0 \ \text{ if } \ n \ge 14, \end{aligned}$$

and

$$\begin{aligned} & f^+(x_{n+1}) - f^+(x_m) \\ = & (2m+3n+1) - (3m+n-1) \\ = & 2n-m+2 > 0 \ \text{ for } \ m \le 2n+1. \end{aligned}$$

Then $f^+(x_{m+1}) = \frac{(1+n)n}{2}$ is the maximum vertex sum of $\cup_{i=1}^m S_i' \cup S_n$ and all vertex sums of $\cup_{i=1}^m S_i' \cup S_n = mS_2 \cup S_n$ are pairwise distinct if $n \ge 14$. Thus f^+ is injective.

Case 2. m is even and $n+2 \le m \le 2n$. Let f be a function defined on $E(\cup_{i=1}^m S_i' \cup S_n)$ satisfying

$$\begin{cases} f(E(S_n)) = & \{1, 2, \cdots n-1, 2n\}, \\ f(E(S_i')) = & \{n-1+i, 2m+i\} \\ & for\ i = 1, 2, \cdots, n, \\ f(E(S_i')) = & \{n+i, m+i\} \\ & for\ i = n+1, n+2, \cdots, m. \end{cases}$$

 ISBN：978-986-02-7580-3

Claim: any f satisfying above conditions is an edge labeling of $\cup_{i=1}^{m} S'_i \cup S_n$. First, we see that

$\cup_{i=1}^{n}\{n-1+i\} = \{n, n+1, \cdots, 2n-1\}$,

$\cup_{i=n+1}^{m}\{n+i\} = \{2n+1, 2n+2, \cdots, m+n\}$,

$\cup_{i=n+1}^{m}\{m+i\} = \{m+n+1, m+n+2, \cdots, 2m\}$,

$\cup_{i=1}^{n}\{2m+i\} = \{2m+1, 2m+2, \cdots, 2m+n\}$

are disjoint sets. And

$$\begin{aligned}
& \{1,2,\cdots,2m+n\} \\
= \ & \{1,2,\cdots,n-1,2n\} \cup \{n,n+1,\cdots,2n-1\} \\
& \cup\{2n+1,2n+2,\cdots,m+n\} \\
& \cup\{m+n+1,m+n+2,\cdots,2m\} \\
& \cup\{2m+1,2m+2,\cdots,2m+n\} \\
= \ & \{1,2,\cdots,n-1,2n\} \cup (\cup_{i=1}^{n}\{n-1+i\}) \\
& \cup(\cup_{i=n+1}^{m}\{n+i\}) \cup (\cup_{i=n+1}^{m}\{m+i\}) \\
& \cup(\cup_{i=1}^{n}\{2m+i\}) \\
= \ & \{1,2,\cdots,n-1,2n\} \\
& \cup(\cup_{i=1}^{n}\{n-1+i, 2m+i\}) \\
& \cup(\cup_{i=n+1}^{m}\{n+i, m+i\}) \\
= \ & f(E(S_n)) \cup (\cup_{i=1}^{n} f(E(S'_i))) \\
& \cup(\cup_{i=n+1}^{m} f(E(S'_i))) \\
= \ & \cup_{i=1}^{m} f(E(S'_i)) \cup f(E(S_n)) \\
= \ & f(E(\cup_{i=1}^{m} S'_i \cup S_n)).
\end{aligned}$$

And $|E(\cup_{i=1}^{m} S'_i \cup S_n)| = 2m+n$. We have $f : E(\cup_{i=1}^{m} S'_i \cup S_n) \to \{1,2,\cdots,2m+n\}$ is a bijection. Hence f is an edge labeling of $\cup_{i=1}^{m} S'_i \cup S_n$. Then the induced vertex sum f^+ of f is

$$\begin{cases}
S_n \ \to \ \begin{pmatrix} 1,2,\cdots,n-1,2n \\ \frac{n^2+3n}{2} \end{pmatrix}, \\
S'_i \ \to \ \begin{pmatrix} n-1+i, 2m+i \\ 2m+n-1+2i \end{pmatrix} \\
\qquad for\ i = 1,2,\cdots,n, \\
S'_i \ \to \ \begin{pmatrix} n+i, m+i \\ m+n+2i \end{pmatrix} \\
\qquad for\ i = n+1, n+2, \cdots, m.
\end{cases}$$

Now we show that f^+ is injective. From above, we see that all the endvertices of $\cup_{i=1}^{m} S'_i \cup S_n$ receive vertex sums $1,2,\cdots,2m+n$. Let

$$A = \{f^+(x_1), f^+(x_2), \cdots, f^+(x_n)\}$$
$$= \cup_{i=1}^{n}\{2m+n-1+2i\},$$
$$B = \{f^+(x_{n+1}), f^+(x_{n+2}), \cdots, f^+(x_m)\}$$
$$= \cup_{i=n+1}^{m}\{m+n+2i\}.$$

We have $A \subset O$ or $A \subset E$ and $B \subset O$ or $B \subset E$. And since

$$\begin{aligned}
& f^+(x_{n+1}) - f^+(x_1) \\
= \ & (m+3n+2) - (2m+n+1) \\
= \ & 2n-m+1
\end{aligned}$$

is odd and positive for m is even and $m \le 2n$. Then we have A and B are in different parties of N and

$$f^+(x_m) > \cdots > f^+(x_{n+2}) > f^+(x_{n+1}) > 2m+n,$$
$$f^+(x_n) > \cdots > f^+(x_2) > f^+(x_1) > 2m+n.$$

i.e. all centers $x_1, x_2, \cdots, x_m$ of $S'_1, S'_2, \cdots, S'_m$ receive different vertex sums $> 2m+n$. Now, we consider the vertex sum $f^+(x_{m+1})$ for the center x_{m+1} of star S_n.
Since

$$\begin{aligned}
& f^+(x_{m+1}) - f^+(x_n) \\
= \ & \frac{n^2+3n}{2} - (2m+3n-1) \\
\ge \ & \frac{n^2+3n}{2} - 4n - 3n + 1 \text{ for } m \le 2n \\
= \ & \frac{n^2-11n+2}{2} > 0 \text{ if } n \ge 14,
\end{aligned}$$

and

$$\begin{aligned}
& f^+(x_{m+1}) - f^+(x_m) \\
= \ & \frac{n^2+3n}{2} - (3m+n) \\
\ge \ & \frac{n^2+3n}{2} - 6n - n \text{ for } m \le 2n \\
= \ & \frac{n^2-11n}{2} > 0 \text{ if } n \ge 14.
\end{aligned}$$

Then $f^+(x_{m+1}) = \frac{n^2+3n}{2}$ is the maximum vertex sum of $\cup_{i=1}^{m} S'_i \cup S_n$ and all vertex sums of $\cup_{i=1}^{m} S'_i \cup S_n = mS_2 \cup S_n$ are pairwise distinct if $n \ge 14$. Thus f^+ is injective.

By Cases 1 and 2, $mS_2 \cup S_n$ is antimagic. □

Theorem 2.7 *For $n \ge 14$, the star forest $mS_2 \cup S_n$ is antimagic if and only if $m \le 2n+1$.*

Proof. $\Rightarrow$) By Lemma 2.3.
$\Leftarrow$) By Lemmas 2.5 and 2.6. □

In fact, for $3 \le n \le 13$, we have found antimagic labelings for all $mS_2 \cup S_n$ whenever $m \le \frac{2n-5+\sqrt{8n^2-24n+17}}{2}$. Then by Lemma 2.3, Remark 2.4 and Theorem 2.7, we have the following theorem:

 ＩＳＢＮ：978-986-02-7580-3

Theorem 2.8 *For $n \geq 3$, the star forest $mS_2 \cup S_n$ is antimagic if and only if*

$$m \leq \min\{2n+1, \frac{2n-5+\sqrt{8n^2-24n+17}}{2}\}.$$

□

3 Conclusion

While we have obtained the above results on star forests, it will be interesting for finding how many S_2 stars can be contained in a S_1-free star forest and the star forest is still antimagic. Now we conclude this paper with the following open problem: for $r \geq 2$, what's the best upper bound of m for $mS_2 \cup (\cup_{i=1}^{r} S_{k_i})$ where $k_i \geq 3$ for $i = 1, 2, \cdots, r$ being antimagic?

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